

YASH GHONGADE

Data Scientist | Machine Learning & Gen AI | RAG | LLMs | Python | Azure | AWS
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PROFESSIONAL SUMMARY

Software Developer skilled in Python, Machine Learning (prediction models), and AI projects, with hands-on experience in Machine Learning using Scikit-Learn Pipelines and Joblib. Proficient in NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, and with strong knowledge of Docker images, Docker containers, and Docker Hub along with that also have the knowledge of RAG and LLMs and MCP. Experienced in data collection techniques and practical Deep Learning concepts including Neural Networks. Also skilled in Web Development and eager to contribute and grow in the tech industry.

TECHNICAL SKILLS

Programming & Tools: Python, SQL, C, Java (Basics), Flask, Django, FastAPI, Web Development, Linux, Jupyter Notebook, Google Colab, Quadratic AI, Tableau, Microsoft SQL Server.

Data Science & Analytics: Pandas, NumPy, Scikit-Learn, Power BI, EDA, Feature Engineering, Matplotlib, Seaborn, Plotly

Machine Learning: Regression, Decision Trees, Random Forest, K-Means Clustering, Cross-Validation, RMSE, MAE, Scikit-Learn Pipelines, Feature Scaling, One-Hot Encoding, Stratified Shuffle Split.

Deep Learning: Neural Networks (ANN), TensorFlow, Keras, PyTorch, Model Fine-Tuning

Generative AI: Prompt Engineering, RAG, Agentic RAG, Re-ranking, Metadata Filtering, Embeddings, GPT, Claude, CodeX, Gemini, OpenAI API, Ollama, Local LLMs, FAISS, Chroma, Azure AI Search, AWS OpenSearch, Hugging Face, Semantic Search.

Agentic AI & Frameworks: LangChain, LangGraph, CrewAI, MCP, Multi-Agent Workflows, Tool & API Integration, Agent Orchestration

DevOps, Cloud & Databases: Docker, Terraform (IaC), Git, GitHub, GitLab, Kubernetes (Basics), CI/CD, MLOps, Azure, AWS (Bedrock, S3), Azure Blob Storage, MySQL, MongoDB

EXPERIENCE

PLS (Process Legal Support) — Python AI Developer, Legal Data Intelligence
New York Legal Technology | Apr 2026 – Present

- Developed production-grade AI-powered legal data extraction and document processing systems using Python, FastAPI, React, Azure AI Document Intelligence (OCR), RAG, PyMuPDF, Pandas, and SQL for New York court workflows.
- Architected Producer-Consumer ETL pipelines to acquire and process thousands of legal documents from NYSCEF and Web Civil, converting unstructured PDF and ZIP files into structured, analysis-ready legal datasets.
- Built automated workflows for case information extraction, ZIP-code identification, LDS server-code assignment, manual corrections, Excel/CSV exports, real-time progress tracking, and stable GUID mapping.
- Integrated LDS REST APIs, SQL Server, MySQL, FTP, and ASP.NET Web API to synchronize legal records including Legal Delivery, Server Files, and File Assignments, with robust retry, timeout, exception handling, and data-consistency controls.
- Developed a secure file management and printing portal using Next.js, TypeScript, React, REST APIs, and ASP.NET Web API, implementing authentication, server-specific file isolation, file assignment management, and administrative controls.
- Engineered API and database integration workflows with automated authentication/API testing, production error handling, retry mechanisms, and legacy-system integration to improve reliability, maintainability, and scalability.
- Built an AI-assisted geospatial territory intelligence application using Python, MySQL, LDS REST APIs, ZIP-code mapping, and interactive visualization, enabling date-range filtering, previous-week analysis, server-code filtering, and territory-level intelligence.
- Implemented idempotent database updates, latest-data detection, synchronized API/MySQL storage, and all-server or selected-server analysis to maintain reliable and consistent territory datasets.
- Engineered scalable web-scraping and data acquisition pipelines using Selenium, BeautifulSoup, Requests, browser session management, persistent Cloudflare sessions, cookies, headers, and request-context reuse to improve large-scale document collection reliability.
- Developed real-time HTML/JavaScript monitoring dashboards for pipeline progress, processing metrics, errors, and failures, enabling improved operational visibility and troubleshooting.
- Automated deployment of Python workflows as standalone Windows executables using PyInstaller, enabling non-technical users to execute data-processing and automation workflows without a Python environment.
- Delivered structured Excel/CSV reporting and data-export workflows for downstream legal operations, supporting efficient analysis and operational decision-making.

- Built end-to-end Machine Learning pipelines using Python, Scikit-Learn, Pandas, NumPy, including data preprocessing, feature engineering, and model training, with evaluation using RMSE, MAE.
- Performed Exploratory Data Analysis (EDA) using Matplotlib, Seaborn to extract actionable insights and improve data-driven decision making.
- Developed and deployed ML & Deep Learning models (Linear Regression, Random Forest, Decision Tree, TensorFlow, Keras) for prediction and classification tasks.
- Designed interactive data applications using Streamlit and worked on RAG-based systems using FAISS, Sentence Transformers, Chunking, Vectorization, LLMs, Prompt Engineering.
- Implemented data collection & processing pipelines using Web Scraping (BeautifulSoup, Requests), SQL (ETL, Joins, Window Functions, Joins, Trigger, View, Procedure), and ensured deployment with Docker, Git, GitHub.

PROJECTS

RAG-Based AI Teaching Assistant (Video RAG) [RAG Project Link](#)

- Converted video/audio to text using OpenAI Whisper generated vector embeddings stored in FAISS for semantic similarity retrieval.
- Built a full RAG pipeline (retrieval + generation) integrating OpenAI API for hallucination-free, context-aware question answering with quality evaluation metrics.

MCP-Based AI Recruitment Screening & Shortlisting Assistant [MCP Project Link](#)

- Designed AI recruitment automation using Model Context Protocol - inspired modular architecture built Flask + MySQL backend with resume upload, PDF parsing (PyMuPDF), and skill-based candidate scoring.
- Integrated SMTP email automation for automated shortlist/rejection notifications. **Stack:** Python, Flask, MySQL, LLMs, Prompt Engineering, REST APIs.

Data Collection & Web Scraping Pipeline [Scraping Project Link](#)

- Scraped and extracted structured data from multiple websites using BeautifulSoup & Requests optimized workflow by caching HTML locally to reduce server hits and improve efficiency.

EDUCATION

B.E. Information Technology — SKNCOE, Pune | 2021–2025 | CGPA: **7.89**

ACHIEVEMENTS & CERTIFICATIONS

- **4th** Place – MIT Gen AI Hackathon
- **3rd** Place – Web Development Hackathon
- Microsoft Azure AZ-900
- Data-Science